TraceErase: How the App Works

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1. What TraceErase is

TraceErase helps an ordinary person find every company that holds their personal data, make those companies delete it under India's Digital Personal Data Protection (DPDP) Act 2023, and keep proof that they did. Its tagline says the whole idea: **Find it. Delete it. Prove it.**

The problem it solves. When you press "delete my account" in an app, three things go wrong:

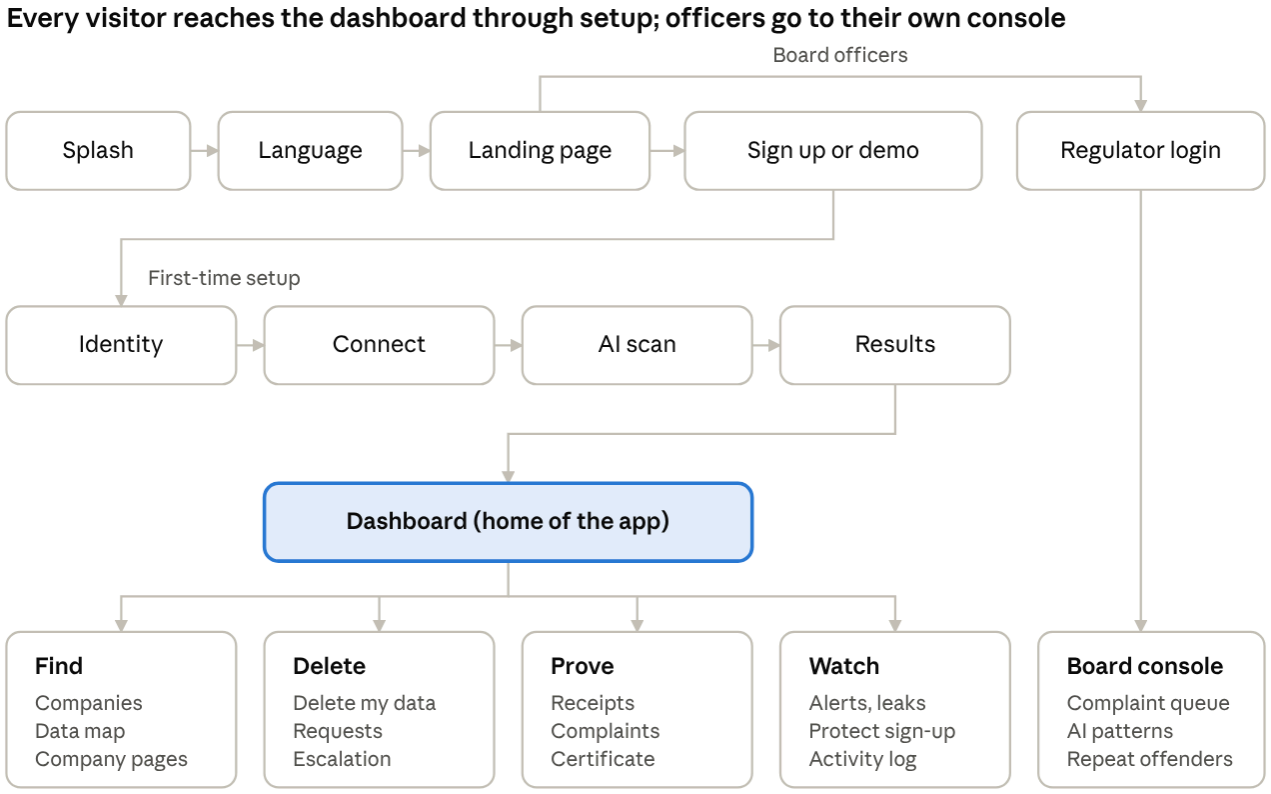
- **Copies stay with partners.** The app may have shared your details with payment, delivery and advertising companies. Your request never reaches them.
- **"Deleted" is just an email.** There is no record you can check, show a court, or hold a company to.
- **Leaks show up months later.** Spam from companies you never signed up with, and no way to know who passed your details on.

The four promises.

Step	What TraceErase does	Where you see it
Find	AI reads who emails you (sender and subject only) and lists every company that has your data, what it holds and who it shares with	AI scan, Results, Companies, Data map
Protect	Each app can get its own protected email address, so a leak points straight at the company that leaked it	Protect a sign-up, Leak page
Delete	One request, signed by you, goes to every chosen company and to every partner a company names in its reply. Reminders, grievances and complaints follow automatically	Delete my data, Deletion requests
Prove	AI checks every reply. Every step gets a sealed receipt anyone can verify. A certificate sums up the result	Evidence and proof, Verify a receipt, Certificate

Who can use it. Common people (real accounts), judges and testers (demo accounts with sample data), and Data Protection Board officers (the regulator console). The public can also see company scores and verify receipts without an account.

Where it runs. The website is at trace-erase.web.app (Firebase Hosting). Sign-in uses Firebase Authentication. All data is stored in Firestore. A small server on Render runs the AI (Google Gemini), sends letters through Brevo and checks deadlines every few minutes.



app map · how the pages connect

A visitor moves left to right through the opening screens, then through setup to the dashboard, which leads to the four parts of the app; Board officers have their own console, where filed complaints arrive.

2. Who uses it: the four roles

There are four kinds of people in TraceErase, and each sees a different part of the app.

Role	How they get in	What they can do	What they cannot do
Visitor (no account)	Opens trace-erase.web.app	Read the landing page, see Company scores, verify any receipt, change language	See anyone's data
Real user	Sign up with email and password (or Continue with Google), then confirm the email	Everything: scan, see companies, delete, get real letters sent, receipts, complaints, certificate, assistant, settings	Use Demo controls; real letters are never sent to companies found only in sample emails
Demo user	Log in page, then **Try the demo** (no email or password)	The full app with sample data, plus the **Demo controls** panel to move time, make companies reply and simulate leaks, calls and SMS	Send anything real. A yellow banner says "Demo account". The account is deleted after 3 days
Regulator (Data Protection Board officer)	Log in page, then **Regulator login**, with an official email and password	The Regulator console: every filed complaint, its evidence, repeat offenders and an AI-found pattern	See a user's private data beyond what the complaint contains

Demo regulator accounts (shown on the regulator login page): regulator@traceerase.demo / Regulator@2026 and board.officer@traceerase.demo / Board@2026.

Real data and demo data are kept apart. Demo accounts have their own demo clock, get pretend company replies and never send real email. A real account can also press "Use sample emails" to try the flow; letters to those sample companies are simulated too.

3. Opening the app

Every visit starts with the splash screen, then the language screen, then either the landing page (new visitors) or the dashboard (people already signed in).

3.1 Splash screen

- **Purpose:** a short welcome while the app loads and the server wakes up.
- **What you see:** the TraceErase shield logo, the tagline "Find it. Delete it. Prove it.", the line "Securing your session", and three promises: Passkey-signed requests, Tamper-proof receipts, Built on the DPDP Act 2023.
- **What you do:** nothing. After a few seconds it moves on by itself.
- **Note:** the free server sleeps when unused. If it takes longer, a small message at the bottom says "Starting the TraceErase server" (up to a minute).

3.2 Language screen ("Your privacy, in your language")

- **Purpose:** choose the language for every screen and for the legal letters AI writes.
- **Fields:** a grid of 12 languages (English, Hindi, Tamil, Telugu, Kannada, Malayalam, Marathi, Gujarati, Bengali, Arabic, Spanish, French). Tap one and the whole app changes immediately.
- **Next:** press **Continue**. Signed-in people go to the dashboard; everyone else goes to the landing page.
- **Later:** the globe button at the top of every page changes the language at any time.

3.3 Landing page (/welcome)

The public home page explains TraceErase to a first-time visitor, top to bottom:

Section	What it says or does
Top bar	Logo; links How it works, AI features, Company scores, Verify a receipt; language globe; **Log in** and **Get started** (or **Open dashboard** when signed in)
Hero	"You clicked delete. Did they?" with **Get started free** and **See how it works**, and a live-looking example: a deletion request on day 12 of 30 and a "Leak caught" card
The problem	Three cards: copies stay with partners, "deleted" is just an email, leaks show up months later
How it works	Four steps: Find, Protect, Delete, Prove
AI agents	Six AI helpers: Discovery agent, Policy reader, Request writer, Reply checker, Leak investigator, Complaint drafter. Personal details are masked before AI sees them

Section	What it says or does
Audiences	For people (free, private) and for companies (nothing to install, answer by email, public score)
Footer call to action	**Create your free account** and **See company scores**; footer links to Company scores, Verify a receipt, Regulator login

Verify a receipt is in the top bar on purpose: it lets anyone without an account (a company, a lawyer, the Board) check that a TraceErase receipt is genuine.

4. Sign up and log in

A real account takes three steps: create the account, confirm the email, and (optionally) add a passkey. Firebase Authentication handles the password and email check.

4.1 Create your account (/signup)

The left side shows a sample "protected identity" card; the right side holds the form. A step bar shows Account → Verify → Passkey.

Field or button	What to enter	Why
Continue with Google	One click, if Google sign-in is switched on	Skips the password and email check; Google already confirmed the email
Full name	Your real name	It goes on your deletion letters and complaints, so companies can match you
Real email	The email companies know you by	Your login, where confirmations and letter copies arrive
Password	At least 8 characters; the eye button shows or hides it	Your login
Real mobile number (optional)	10-digit Indian number	Used only in legal letters to identify you. SMS checking needs a paid plan, so it is saved without a code
I agree to the Terms and Privacy policy	Tick	Required

Field or button	What to enter	Why
Create account	Press	Creates the account and sends the confirmation email
Try the demo	Press instead of signing up	Opens a separate demo account with sample data (see section 11)

4.2 Confirm your email (/signup/check-email)

- A designed TraceErase email arrives with a **Confirm my email** button (check Spam if missing).
- Click it, come back, and press **Continue**. The page also checks by itself when you return to the tab.
- **Send the email again** is available after 30 seconds. **Wrong email? Start again** goes back to sign-up.

4.3 Create your passkey (/passkey)

- **Purpose:** your fingerprint, face or device PIN becomes your signature on every deletion request, so nobody can send one in your name.
- **What you do:** press **Create passkey** and follow your device's prompt. Nothing leaves your device.
- **Skip for now:** allowed. Requests are then signed with your account instead, and you can add a passkey later in Settings → Security.

4.4 Log in (/login)

Three ways in, plus two extra doors:

1. **Log in with passkey** — the fastest; one touch.
2. **Continue with Google** — when switched on.
3. **Email and password** — with **Forgot password?**, which sends a designed reset email.
4. **Try the demo** — a fresh demo account, no details needed.
5. **Regulator login** — the link at the bottom, for Data Protection Board officers.

On a computer running TraceErase without Firebase (local development), the log in page instead offers a one-time code by email or SMS, entered on the **Enter the codes** page.

4.5 Regulator log-in (/regulator/login)

- Fields: **Official email** and **Password**; the demo accounts are listed on the page.

- After logging in, the officer lands on the Regulator console (section 12).

5. First-time setup (onboarding)

After sign-up, four short screens find out who has your data. A progress bar shows Identity → Connect → Scan → Results, and **Skip for now** (top right) goes straight to the dashboard.

5.1 Identity: "Meet your protected identity"

- **Purpose:** introduces the idea of protected details: a separate email address per app, and one protected phone number, so your real ones stay private.
- **What you see:** your protected number (demo accounts) or the note that phone protection needs a paid phone service; an example protected email for Zomato; and "How a message reaches you" (company → protected address → TraceErase checks it → your real inbox).
- **Tip on the page:** keep using your real number for banks, UPI and government services.
- **Next:** press **Continue**.

5.2 Connect: "Connect what AI should scan"

You choose where AI looks for companies. Pick one card, then press **Start AI scan**.

Card	What it does	Status today
Gmail	AI reads only who emailed you and the subject line, never message bodies	Waiting for Google approval (Google must review apps that read Gmail)
Upload your email export	Upload a Google Takeout .mbox file (or a CSV of sender, subject, date). It is read inside your browser; only sender, subject and date are used. The card lists the three Takeout steps	Works now
Use sample emails	One click: a realistic sample mailbox of 43 senders (Zomato, Myntra, MakeMyTrip and more) plus sample Google sign-ins is loaded and scanned. Letters to these sample companies are simulated	Works now; selected by default

Two boxes underneath state the rules: **What AI reads** (sender names and addresses, subject lines, sign-in app names) and **What we never do** (read message bodies, send email as you, sell or share your data, or train AI on it).

5.3 Scan: "Finding everyone who has your data"

- **Purpose:** shows the Discovery agent working, live.
- **Left:** four sources with progress: the mailbox (emails checked), Google sign-ins (apps listed), AI identifying companies, and reading what each company keeps.
- **Centre:** a ring with the number of companies found and the percentage done.
- **Right:** "Just found", with companies appearing as AI finds them.
- **Bottom:** "Your name, email and number are masked before any AI sees them" and the **See results** button, enabled when the scan finishes.

5.4 Results: "N companies have your data"

- **Summary panel:** the count, an AI-written headline (for example which companies share with the most partners), and four boxes that also work as filters: Not used in a year, Share with partners, Need a web form, Categories.
- **Show filters:** All, Not used in a year, Used this month, Share with partners, Need a web form, each with its count. **Select these** ticks every company in the current filter.

- **Category tabs:** Food, Shopping, Travel, Finance, Health and so on.
- **Company cards:** name, category, when it last emailed you, the kinds of data it holds (Email, Phone, Address, Payment…), how many kinds of partners it shares with, and whether it takes requests by email or web form. Tick a card to select it.
- **Bottom bar:** how many are selected, **Decide later** (go to the dashboard) or **Delete from selected** (start a deletion request with them).

6. The main app

Once set up, every page shares one frame: a navy sidebar on the left and a top bar.

Sidebar. Overview: Dashboard, Data map, Companies (with a count), Alerts (with an unread count). Take action: Deletion requests, Protect a sign-up, Evidence and proof, Ask TraceErase, Activity log, Settings. At the bottom, the protected-number card and **New protected email**.

Top bar. Search companies, **Delete my data** (always one click away), the language globe, the alerts bell and your initials (opens Settings). Demo accounts also see a yellow "Demo account" banner and the **Demo controls** button at the bottom right.

6.1 Dashboard (/app)

- **Greeting:** Good morning / afternoon / evening, with your name.
- **Four tiles:** Companies with your data (and how many were found this month), Being deleted (with a progress bar), Deleted with proof (signed receipts), Leaks caught. Each tile opens its page.
- **Leak banner:** when a leak is caught, a red line names who emailed which protected address and which company AI traced it to, with **Investigate**.
- **Current request card:** the latest deletion request, how many companies it covers and its day, with **Open request**; or **Choose what to delete** if none yet.
- **Where your data lives:** a small preview of the data map, with **Open data map**.
- **Privacy score (out of 100):** based on unused accounts, real emails still in use, leaks and ignored requests. Higher is better.
- **Needs you:** the alerts that need an action; "Nothing needs you right now" when clear.
- **Today's summary:** a short AI summary, and a shortcut to Ask TraceErase.
- **No companies yet?** A card offers **Start the scan**.

6.2 Data map (/app/map)

- **Purpose:** a picture of where your data went: You → companies → their partners → leaks.

- **Columns: Companies** (12 at first, **Show all companies** for the rest), **Their partners** (solid lines = partners a company named in its own reply; dashed "Likely" boxes = kinds of partners its privacy policy says it shares with, such as payment processors or advertising partners), and **Never given** (red, for leaks).
- **Colours:** grey not asked, blue says deleted, brown partly deleted, green verified, orange ignored your deletion, red dashed a leak.
- **Filters:** Everything, Only shared, Leaks.
- **Click a company:** its lines light up and the right panel shows its status, what it shared with whom, its one-line policy summary and **Open**. **Click a "Likely" partner kind:** the panel lists which of your companies share with that kind, with **Ask them to delete**.

6.3 Companies (/app/companies)

- **Purpose:** the full list of every company that holds your data.
- **Filters:** All, Shared your data, Unused 1+ year, Using real email, Asked to delete, Leaks; plus **Find a company**.
- **Table columns:** Company, Your data (what it holds), Shared with, How we reach them (email or web form), Last used, Status. 12 per page.
- **Actions:** tick companies and press **Delete selected (n)**; **Protect a sign-up**; open any company.

6.4 A company's page (/app/companies/:id)

- **Header:** name, category, when it last contacted you, and buttons **Ask about it**, **Fix old account** and **Delete my data**.
- **Protected email card** (when it has one): the address, leaks on it, **Copy**, who is allowed or blocked, and **Rotate this address** (a new address; the old one stops working). If it still has your real email, **Switch to a protected email**.
- **Policy card:** an AI summary of the company's privacy policy: what it collects, how fast it deletes, what it must keep by law (for example tax invoices), who it shares with, whether it sells data, and its privacy contact. It says whether this came from the published policy, AI general knowledge, or general rules, and has **Read the policy again**.
- **Deletion result:** what was deleted, what was kept and why, and who it shared your data with, each with its signed receipt; **See on data map**.
- **History:** every event for this company, newest first.
- **Public score:** its grade, % deleted on time and leaks traced (marked "sample" when based on illustrative figures).

6.5 Fix old account (/app/companies/:id/fix)

For an account you may want to keep. It explains the situation (last used, still has your real email, shares with n kinds of partners), then three steps:

1. **Change your email on the company's site** to a protected address (**Copy**, **Open site**, then mark it switched).
2. **Ask what the company holds** under Section 11 of the DPDP Act: AI writes the request (From, Subject, Request text) and it is sent; the reply comes back and AI reads it.
3. **Decide: Skip and delete now** (starts a deletion request) or **Keep the account** (it stays, now on a protected email, and any leak is caught).

A side box, "Being honest about limits", lists what the company likely holds and which partners are known or unknown.

7. Deleting your data

One signed request goes to every company you choose; TraceErase then sends the letters, reads the replies, follows up and escalates on its own.

7.1 Delete my data: choose (/app/delete)

- **Step bar:** Choose → Review → Sign.
- **Company list:** every company with a tick box, when you last used it, and how it takes requests (Email, written by AI, or Web form, text prepared for you). Partners a company already named are **included automatically** under it.
- **Tools: Find a company**, **Select unused (n)** (everything not used for a year), **Clear**.
- **Summary panel:** how many companies (you chose + partners), how many by email and by web form, the legal deadline (30 days), and an AI note: it writes each request in the format that company accepts and checks every reply for partners it names.
- **Next: Review and sign**.

7.2 Sign your deletion request (pop-up)

- Lists every company in the request, with the channel and anything it may keep by law (for example tax invoices).
- **Handle verification codes** (ticked): when a company emails a code to confirm your deletion request, TraceErase fills it in. Only for these requests, never logins or payments.
- **Escalate if ignored** (ticked): reminder on day 7, formal grievance on day 10, complaint ready on day 15.

- A fingerprint of exactly this request is shown; your passkey signs it (or your account, if you have no passkey).
- **Confirm and send** sends it; **Cancel** goes back.

7.3 What happens after you press Confirm and send

1. AI writes one letter per company, in your letter language, citing Section 12(3) of the DPDP Act, with a reference number (DR-0001…).
2. Each letter goes to the company's privacy or grievance address found in its policy (or its standard privacy@ address). It is sent automatically as "Your name via TraceErase", with a copy to your own email; replies come to your inbox.
3. Web-form-only companies get the text ready to paste into their form.
4. A sealed receipt records that you signed the request; the 30-day legal clock starts.
5. Every step appears in the request page, the alerts and the activity log.

Letters to demo companies, and to companies found only in sample emails, are simulated: written and recorded, but never really sent.

7.4 Deletion requests (/app/requests) and a request's page

- **Deletion requests** lists every request: its number, when it was signed, its day, and how many companies replied; **New request** starts another.
- **A request's page** shows "Deleting your data from N companies", the day (day X of 30), a ring of how many are resolved, counts by status, and the **Automatic escalation** track: Sent (day 0) → Reminder (day 7) → Grievance (day 10) → Complaint ready (day 15) → Board decides.
- **Banners:** "Sent automatically" for real letters; for sample data, a **Get replies now** button that makes the sample companies answer at once; a warning when a company ignored you, with **Review complaint**.
- **Table:** Company (and "partner of X"), Sent by, Latest event, Result, Proof (Receipt or Evidence link), and an arrow to the company's page.
- **Issue certificate / Certificate:** creates the deletion certificate (section 9).

What each result means:

Status	Meaning
AI is writing the letter	The letter is being written and sent
Waiting for reply	Sent; the company has not answered yet

Status	Meaning
Web form / Submit web form	This company only accepts its own form; paste the prepared text there
No response	No reply after the day-7 reminder
Overdue	No reply after day 15; a complaint is ready
Refused	The company said no
Says deleted	The company confirmed deletion; AI checked the reply
Partly deleted	Deleted everything except what a law requires it to keep, with the law named
Verified	Confirmed in a signed email, then nobody used the address it knew you by for 30 days. Strongest proof

7.5 One company inside a request (/app/requests/:id/targets/:tid)

- **Header:** company name, status, partner of whom, day, and which identifier it knows you by; **Ask about this**.
- **Timeline:** every event with its time: letter sent, verification code filled, reminder, grievance, reply, overdue; each links to its receipt.
- **AI reply check:** Genuine reply / Needs a look / Not genuine, with checks (signed by the company's domain, gives a reference number, says the data was deleted, names partners) and a one-line summary.
- **Result:** what was deleted and what was kept (with the law), plus the signed receipt.
- **Partners named in the reply:** each one automatically gets its own deletion request.
- **Got a reply in your inbox?** Paste the company's email here and press **Check the reply**; AI reads it exactly as above.
- **The letter:** subject, body, recipient, when it was sent, and **Copy**. If automatic sending failed, a card with the address and **Send** lets you retry. For web-form companies: **Open site**, a ticket number box, and **I've submitted it**.
- **After a confirmation:** "We keep watching your X address. Any message now would be a leak", and a countdown until it becomes Verified.

8. Protection, alerts and leaks

Protected addresses turn every leak into evidence: if a stranger emails an address you gave only to one company, that company (or a partner) leaked it.

8.1 Protect a sign-up (/app/protect)

- **Purpose:** before signing up for a new app, get a protected email just for it.
- **What you do:** type the app or website in **Which app or website?** (or tap a suggestion under **Try:**) and press **Generate**.
- **Result card "Ready for X":** the new address (and the protected number, where available) with **Copy**, and **What happens next**: use these details to sign up; their emails reach your inbox; anyone else is caught as a leak.
- **Your protected emails:** a table of every address (App, Address, Status: Healthy, n leaks, or Retired) with **Manage**.
- **Manage pop-up:** the allowed senders and blocked senders for that address, a box to add an **Allowed sender** (for example swiggy.com), **Remove**, **View company**, and **Rotate this address** (the old one stops working).
- **Availability:** works fully in demo accounts. Real accounts need TraceErase's own email domain to receive mail, which the free setup does not have yet, so the page explains this.

8.2 Alerts (/app/alerts)

- **Purpose:** everything that happened and what to do next.
- **Tabs:** All, Needs action, Leaks, Replies, Complaints, Sharing, each with a count. Alerts are grouped by day.
- **Each alert:** an icon, a title (for example "Leak: PizzaPoint emailed your Zomato address"), a short explanation, the time, and an **Open** button to the right page. **Mark all as read** clears the badges.
- **Alert me about** (switches): Leaks, Company replies, Deadlines and escalations, New companies found.
- **Send alerts by** (switches): In the app, Email, SMS (works on weak internet).
- **Looking for something older?** Links to the activity log.

8.3 A leak (/app/leaks/:id)

- **Header:** "Leak detected: {sender} got the email you gave only to {company}".
- **AI investigation:** how the sender likely got your address, with a confidence (High / Medium / Low), a chain picture (You → company → partner → sender), and "Why {name}": for example, the company itself said in writing that it shared your data with an ad network. **Run the investigation again** re-checks.
- **The message we stopped:** Blocked, From, Subject.
- **What to do** (one click each):
  1. **Blocked {sender} only** — done automatically.

2. **Ask {sender} to delete it** — AI writes a deletion request, signed by you.
3. **Give {company} a new address** — rotates the address; the old one stops working.
4. **Save the evidence** — builds a timestamped evidence pack.
5. **Add to complaint** — puts the leak into a complaint to the Board.

- **Who may use this address:** the allowed senders, with a link to manage protected addresses.

9. Proof: receipts, evidence, complaints and certificates

Every important step gets a sealed receipt, so you can later prove what happened and when to a company, a lawyer or the Data Protection Board.

How sealing works, in simple words. TraceErase turns each record into a unique code, like a fingerprint; change one letter and the code changes completely. Every minute, the new codes are sealed together, and each seal also includes the seal before it, so an old record cannot be quietly changed later. Only the code is sealed, never your personal details.

9.1 Evidence and proof (/app/evidence)

- **Three tiles:** Receipts (proof of each request, reply, sharing and leak), Evidence packs (ready for complaints), Complaints (and how many were filed).
- **Complaints, packs and certificates:** each complaint (draft or filed, and against whom), each evidence pack (how many records), each certificate (how many companies resolved).
- **Receipts list,** with a short key: **Deletion request signed** (proof you asked, and when; the 30-day deadline counts from here), **Company reply** (what the company said, deleted or kept), **Deletion certificate** (one summary of the whole request), **Sealed** (locked so it cannot be changed). Each receipt shows Sealed or Sealing…

9.2 A receipt (/app/receipts/:id)

- **Left:** what the receipt is (for example "Zomato confirmed deletion"), Genuine / Changed / Being sealed, when it was issued, its source (for example a DKIM-signed email from zomato.com), what was deleted, what was kept by law, the request number, the company's reference, the receipt ID. For a sharing receipt, a picture: From company → the data → To partner, with the purpose and how long it may be kept.
- **Buttons: Use as evidence** (builds an evidence pack) and **Public verify page**.
- **Right, "Tamper-proof seal":** the record's unique code, which seal holds it, when it was sealed, the seal's code, and three checks: not changed, inside its seal, seals unbroken.

- **What this means:** this receipt proves the step happened and when; if anyone changes it, the verify page shows it.

9.3 Evidence pack (/app/evidence/:id)

- A bundle of records for one company or leak: each record with its source, time and fingerprint, and "Together, this proves" in plain words.
- **Save as PDF**, **Use in a complaint**, and **Verify a record**.
- Packs are built automatically when a company ignores you or a leak is caught, or by you from a receipt or leak.

9.4 Complaint to the Data Protection Board (/app/complaints/:id)

- **Purpose:** when a company ignores you or leaks your data, AI drafts a complaint under the DPDP Act 2023 from your evidence. Nothing is filed until you approve, unless you switched on automatic filing.
- **Contents:** Against (the companies), What happened (AI-drafted, editable), I ask the Board to (the remedies), the request it relates to, and the evidence pack attached (n records, all fingerprinted).
- **To file:** tick "I confirm these facts are true and want this complaint filed in my name" and press **File complaint**. It gets a reference (for example DPB-2026-8801) and appears in the Regulator console.
- **Save as PDF** and **What happens next** explain the Board's process.

9.5 Deletion certificate (/app/certificates/:id)

- One page summing up a whole request: number, issue date, and a row per company with its Result (Verified, Says deleted, Partly deleted, Overdue…), Kept and why, and the Proof.
- **Save as PDF** (with a QR code), **Public verify link**, and the certificate's fingerprint. Any change to it breaks the check.

9.6 Verify a receipt (public, /verify)

- Anyone can paste a receipt ID (or scan the certificate's QR code) and press **Verify**; no account needed.
- **Result:** "Genuine. This record has not been changed since it was created", or a warning if it was changed. It shows what the record is, when it was created, its unique code, which seal holds it, the seal's code, where it is kept, and the three checks.
- **How this check works** explains fingerprints and seals in three short lines.

10. Ask TraceErase, the activity log and settings

10.1 Ask TraceErase (/app/assistant)

- **Purpose:** ask any question about your own data in plain language, for example "How did PizzaPoint get my email?" or "When is Zomato's deadline?".
- **How it answers:** Gemini reads only your TraceErase records (receipts, requests, alerts), with your name, email and phone replaced by placeholders, and answers with the facts it used. If Gemini is down, built-in rules answer instead, and the page says so.
- **What you do:** type in **Ask anything about your data**, or press **Speak** to use your voice, then **Send**. Questions can be in your language.
- **Left list: New question** and your recent questions.
- **Shortcuts:** "Ask about this" buttons on company, request and leak pages open the assistant with the question filled in.

10.2 Activity log (/app/activity)

Everything you did and everything TraceErase did for you, with the time. Two tabs:

- **Your activity:** grouped by day; each entry has an icon, a title, a short detail, the time and who did it (**You** or **TraceErase**). It records sign-ins and sign-outs, settings changes (and which setting), scans, deletion requests prepared and signed, each letter sent, each company reply, reminders, grievances, leaks, protected addresses, evidence packs, complaints and certificates. Filters by type (Sign-ins, Scans, Requests, Letters, Replies, Escalations, Protection, Leaks, Proof, Alerts) and by who (Everyone, By you, By TraceErase). Click an entry to open its page.
- **Messages to your protected details:** every email, call and SMS that reached a protected address or number: when, from whom, to which address, what TraceErase did (Forwarded, Blocked, Code filled, Rang through) and why. Tiles count forwarded messages this week, blocked messages (and how many were leaks) and codes handled. The sender is logged, never the message content.

10.3 Settings (/app/settings)

A side menu jumps to each section:

Section	What is in it
Profile	Name (editable, **Save**), member since, real email and real number (marked private)

Section	What is in it
Protected identity	Protected number (or why it is not available), protected emails count and leaks caught, **Manage**
Automation	Three switches: fill verification codes for my deletion requests; send reminders and grievances automatically (day 7 and day 10); file complaints automatically when a company ignores everything
AI (Google Gemini)	Whether AI is working, the model in use, the model for quick tasks, the service, when it last answered, **Test again**, and why it is not working if not. Personal details are always masked before AI
Language and display	App language; letters to companies (the language AI writes your requests in); text size (Small, Default, Large); theme (Light, Dark, System)
Security	Your passkeys (added, last used, **Remove**), **Add a passkey**, **Log out**
Your TraceErase data	**Export** everything we hold in one file; **Delete account** (asks you to confirm, then removes everything)

11. Demo controls (demo accounts only)

The **Demo controls** button (bottom right, demo accounts only) opens a side panel that lets a judge see 30 days of TraceErase in two minutes. It never touches real companies, and each demo account has its own demo clock, so it never changes anyone else's dates. The top of the panel shows whether Gemini is working, with **Test Gemini**.

Story tab: run the full story in order

Step	Button	What you will see
1	Set up demo story	Zomato and MakeMyTrip get protected emails
2	(Delete my data)	Choose companies and sign a request, as a real user would
3	Reply now (Companies reply)	Zomato replies, deletes your account, keeps tax invoices by law, and names Razorpay and AdNexa as partners; both get their own requests automatically. Razorpay replies too

Step	Button	What you will see
4	Move 15 days ahead	Reminders on day 7, MakeMyTrip answers after its reminder, a grievance on day 10; AdNexa ignored everything, so on day 15 a complaint is prepared
5	Simulate a leak	PizzaPoint emails your Zomato-only address; it is blocked and AI traces it to AdNexa
6	Move 30 days ahead	MakeMyTrip's address stayed silent for 30 days, so it becomes Verified

Messages tab: test the protected identity

- **Send an email to a protected address:** choose the address, From, Subject, Message, and whether it has a valid DKIM signature (untick to test a fake email). **Deliver email** shows it being forwarded, blocked, or caught as a leak.
- **Send an SMS to the protected number:** From and text; codes for your requests are filled automatically.
- **Simulate a call:** Name and Phone; apps you use ring through, strangers are blocked.

Clock and tools tab

- **Demo clock:** move forward by a chosen number of days to see reminders, grievances, complaints and verification; **Reset clock to today**.
- **Activity log:** jump to the log.
- **Anchor receipts:** seal the newest receipts now instead of waiting a minute.

Real accounts that used **Use sample emails** have no panel, but their request page shows **Get replies now**, which makes those sample companies reply at once.

12. Public pages and the regulator console

12.1 Company scores (/scores, public)

- **Purpose:** a public transparency board: who actually deletes your data?
- **Top:** totals across all TraceErase users: requests tracked, % proven deleted, % ignored, leaks traced.
- **Search a company** (for example Swiggy).
- **Company ranking table:** Company, Grade (A+ to F), Deleted on time (%), Avg. days to delete, Leaks traced, Proof type (signed email, silence-verified, web form), and Data (Live or Sample).

- **How grades are made:** on-time deletion, strength of proof and leaks traced. A company needs at least 3 real requests to get a live score; until then, rows marked **Sample** use illustrative figures and say so.

12.2 Verify a receipt (/verify, public)

See section 9.6: anyone pastes a receipt ID and sees whether the record is genuine and unchanged.

12.3 Regulator console (/regulator, Data Protection Board officers)

- **Header:** "Data Protection Board · prototype console", who is signed in, and **Log out**.
- **Four tiles:** complaints filed, complaints with sealed evidence, companies named, leaks traced.
- **AI found a pattern across complaints:** a headline (for example "AdNexa: 1 ignored deletion and 1 traced leak"), a short explanation, a priority (High / Medium / Low) and a suggested action (for example "Merge into one case").
- **Complaint queue:** Reference, Against, Issue, Evidence (how many verified records), Filed date. Every complaint arrives with fingerprinted evidence, ready to act on.
- **Repeat offenders:** companies ranked by complaints, ignored deletions and leaks, with how the score is built.

13. Behind the scenes

13.1 The AI agents (Google Gemini)

Agent	What it does	Where you see it
Discovery agent	Groups senders by company, names them, sorts them into categories and estimates what data each holds	AI scan, Results
Policy reader	Downloads each company's privacy policy and explains what it keeps, for how long, and who it shares with; finds the privacy contact	Company page, letters
Request writer	Writes legally correct deletion, reminder, grievance and access letters in your letter language	Every letter

Agent	What it does	Where you see it
Reply checker	Decides whether a reply really confirms deletion, lists what was kept and why, and extracts named partners	Request and company pages
Leak investigator	Traces which company passed on an address a stranger used, with a confidence level	Leak page
Complaint drafter	Writes the complaint to the Data Protection Board from the evidence	Complaint page
Assistant and summaries	Answers questions, writes the dashboard summary, translates text not yet translated, and spots patterns for the regulator	Ask TraceErase, Dashboard, Regulator console

If Gemini is unavailable, built-in rules take over and pages are labelled accordingly, so the app keeps working.

13.2 Privacy rules

- Your name, email and phone are replaced with placeholders before anything reaches AI.
- AI reads only sender names, sender addresses and subject lines, never message bodies.
- Only fingerprints are sealed, never personal details. You can export or delete everything from Settings.

13.3 The timeline rules

Day	What happens automatically
0	You sign; letters go out; the 30-day legal clock starts
7	A reminder to companies that have not replied
10	A formal grievance to their grievance officer

Day	What happens automatically
15	The company is marked Overdue; an evidence pack and a complaint are prepared (filed automatically if you allowed it)
30	The legal deadline. An address that stayed silent for 30 days after a company confirmed deletion becomes Verified

A scheduled job checks these deadlines every 10 minutes, and keeps the free server awake.

13.4 What is real and what is simulated on the free setup

Feature	Real accounts	Demo accounts
Sign-in, storage	Real (Firebase Authentication, Firestore)	Real, separate account
AI	Real (Gemini)	Real (Gemini)
Letters to companies	Really emailed through Brevo, copy to you	Simulated
Companies from "Use sample emails"	Letters simulated, replies on request	Simulated
Company replies	Arrive in your inbox; paste them for AI to check	Pretend replies on a schedule
Gmail scan	Waiting for Google's approval; use the email export or sample emails	Sample inbox
Protected emails	Need TraceErase's own email domain (not on the free setup)	Work
Protected phone number and SMS check	Need a paid phone service	Simulated

14. A complete user story: Priya uses every feature

Priya, a college student in Chennai, keeps getting spam from a pizza shop she never signed up with. She tries TraceErase. Her story touches every page in this document, in order. She starts in the demo (so every feature works, including the parts that need a paid service), then repeats the key steps with her real account.

Day 0: first visit

1. **Splash → Language.** She opens trace-erase.web.app, sees the splash, picks **Tamil** on the language screen and presses **Continue**. Every screen is now in Tamil; later she switches to English with the globe button.
2. **Landing page.** She reads "You clicked delete. Did they?", the three problems, the four steps and the six AI agents. She opens **Company scores** and sees Zomato graded B and AdNexa marked as a frequent offender (sample rows are labelled).
3. **Try the demo.** On **Log in**, she presses **Try the demo**. A demo account opens with the yellow "Demo account" banner.

Day 0: setup and scan

4. **Identity.** She meets her protected identity: a protected number (+91 70000 …) and an example Zomato-only email. **Continue**.
5. **Connect.** The Gmail card is already set to the sample mailbox. She presses **Start AI scan**.
6. **Scan.** She watches 43 senders checked, 6 Google sign-ins listed, companies appearing under "Just found", and the ring reaching 39 companies. **See results**.
7. **Results.** The headline says 14 accounts are unused for a year. She presses the **Not used in a year** filter, then **Select these (14)**, and looks at the Food tab out of curiosity.

Day 0: exploring

8. **Dashboard.** Four tiles, privacy score 46 out of 100, Needs you, Today's summary.
9. **Data map.** She clicks Zomato: dashed lines to "Payment processor" and "Advertising partners" (from its policy). She clicks "Advertising partners" and sees five of her apps share with that kind of partner. **Show all companies** fills the map.
10. **Companies.** She filters **Using real email**, opens **Zomato** and reads its policy card: keeps tax invoices for about 8 years, shares with payment and delivery partners.
11. **Fix old account.** For Swiggy, which she still uses, she opens **Fix old account**, copies a protected email to change on Swiggy's site, presses **Write the request** to ask Swiggy what it holds (Section 11), and chooses **Keep the account**.
12. **Protect a sign-up.** She is about to join a new food app, Zepto. She types "zepto.com", presses **Generate**, copies the new protected email, and opens **Manage** to add zepto.com as an allowed sender.
13. **Demo controls → Set up demo story.** Zomato and MakeMyTrip now know her only by protected emails.

Day 0: delete

14. **Delete my data.** She ticks Zomato, MakeMyTrip and Myntra, sees the summary (3 by email, 30-day deadline) and presses **Review and sign**.
15. **Sign.** She keeps **Handle verification codes** and **Escalate if ignored** ticked, sees the request's fingerprint and presses **Confirm and send**, signing with her passkey (she had added one in Settings → Security).
16. **Request page.** Each company shows "AI letter emailed … (demo: simulated)". She opens Zomato's row and reads the AI-written letter citing Section 12(3) of the DPDP Act.

Day 1 to 45: the story unfolds (Demo controls)

17. **Reply now.** Zomato replies: account deleted, tax invoices kept by law (GST), and it names **Razorpay** and **AdNexa** as partners. AI marks Zomato **Partly deleted**, issues a sealed receipt, and sends Razorpay and AdNexa their own requests automatically. Razorpay replies too.
18. **Move 15 days ahead.** A reminder goes out on day 7 and a grievance on day 10; MakeMyTrip answers after its reminder (**Says deleted**). AdNexa ignores everything: on day 15 it turns **Overdue** and a complaint is prepared. An alert says "Complaint ready against AdNexa".
19. **Simulate a leak.** PizzaPoint emails her Zomato-only address. It is blocked, and an alert reads "Leak: PizzaPoint emailed your Zomato address". On the **Leak page**, the AI investigation traces it to **AdNexa** with medium confidence: Zomato said in writing it gave her data to AdNexa. She presses **Ask PizzaPoint to delete it**, **Give Zomato a new address** and **Save the evidence**, then **Add to complaint**.
20. **Messages tab.** She sends a test email from "offers@unknown.example" to her Zepto address with DKIM unticked: blocked as a fake. She sends an SMS with a code: filled automatically. She simulates a call from Swiggy's delivery partner: it rings through.
21. **Move 30 days ahead.** MakeMyTrip's address stayed silent for 30 days: it becomes **Verified**, the strongest proof.

Day 45: proof and justice

22. **Complaint.** She opens the complaint against AdNexa, reads what AI drafted, ticks "I confirm these facts are true" and presses **File complaint**. It gets reference DPB-2026-8801. She saves it as a PDF.
23. **Certificate.** On the request page she presses **Issue certificate**: one page listing Zomato Partly deleted (tax invoices, GST law), Razorpay Partly deleted, MakeMyTrip Verified, Myntra Says deleted, AdNexa Overdue (complaint filed). She saves it as a PDF with its QR code.

24. **Evidence and proof.** She sees her receipts (signed request, each reply, each sharing, the leak), the evidence pack and the complaint. She opens one receipt: "Genuine, not changed", with its seal.
25. **Verify a receipt.** She sends the certificate's public link to her lawyer, who opens it without an account and sees "Genuine. This record has not been changed since it was created".
26. **Ask TraceErase.** She asks by voice, "How did PizzaPoint get my email?" and gets: most likely through AdNexa, which Zomato said in writing received her address.
27. **Alerts and activity log.** She marks all alerts as read and switches off SMS alerts. The activity log shows every step of the last 45 days, by her and by TraceErase.
28. **Settings.** She sets her letter language to Hindi, turns on **File complaints automatically**, switches to the Dark theme, and exports her data.

The regulator's side

29. **Ravi, a Data Protection Board officer,** logs in through **Regulator login**. Priya's complaint is in the queue with its verified records. The AI pattern card says "AdNexa: 1 ignored deletion and 1 traced leak", priority High, suggesting **Merge into one case**. AdNexa tops Repeat offenders.

Then with her real account

30. Priya presses **Create a real account** in the banner, signs up with her email and a password (eye button to check it), opens the designed confirmation email, presses **Confirm my email**, and adds a passkey.
31. On Connect she uploads her Google Takeout file (or presses **Use sample emails** to try first), runs the scan, and sends a real deletion request. Real letters go out automatically as "Priya via TraceErase", with copies in her Gmail.
32. When Zomato's reply lands in her inbox, she pastes it into **Got a reply in your inbox?**; AI checks it, and the receipt, escalation, complaint and certificate steps work exactly as in the demo. After 3 days the demo account deletes itself.